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Original research

Nutrient intake, meal timing and sleep in elite male Australian football players

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ABSTRACT

Objectives: To investigate the relationship between dietary intake, meal timing and sleep in elite male Australian football players.
Design: Prospective cohort study.
Methods: Sleep and dietary intake were assessed in 36 elite male Australian Football League (AFL) players for 10 consecutive days in pre-season. Sleep was examined using wrist activity monitors and sleep diaries. Dietary intake was analysed using the smartphone application MealLogger and FoodWorks. Generalised linear mixed models examined the associations between diet [total daily and evening (>6 pm) energy, protein, carbohydrate, sugar and fat intake] and sleep [total sleep time (TST), sleep efficiency (SE), wake after sleep onset (WASO) and sleep onset latency (SOL)].
Results: Total daily energy intake (MJ) was associated with a longer WASO [$\beta = 3$, 95%CI: 0.2–5; $p = 0.03$] and SOL [$\beta = 5$, 95%CI: 1–9; $p = 0.01$]. Total daily protein intake ($g\,kg^{-1}$) was associated with longer WASO [$\beta = 4$, 95%CI: 0.8–7; $p = 0.01$] and reduced SE [$\beta = -0.7$ CI: −1.3 to −0.2; $p = 0.006$], while evening protein intake ($g\,kg^{-1}$) was associated with shortened SOL [$\beta = -2$, 95%CI: −4 to −0.4), $p = 0.02$]. Evening sugar intake ($g\,kg^{-1}$) was associated with shorter TST [$\beta = -5$, 95%CI: −10 to −0.6; $p = 0.03$] and WASO [$\beta = -1$, 95%CI: −2 to −0.3; $p = 0.005$]. A longer period between the evening meal consumption and bedtime was associated with a shorter TST [$\beta = -8$, 95%CI: −16 to −0.3; $p = 0.04$].
Conclusions: Evening dietary factors, including sugar and protein intake, had the greatest association with sleep in elite male AFL players. Future research manipulating these dietary variables to determine cause and effect relationships, could guide dietary recommendations to improve sleep in athletes.
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Practical implications

To enhance sleep from a nutritional perspective, elite male Australian Rules football players may benefit from:

- Ensuring an adequate intake of food in the hours leading to bedtime.
- Consuming a low sugar meal or snack in the evening.
- Consuming whey-based protein before bed, such as milk or a powered protein supplement.

1. Introduction

The recovery from training and competition is paramount to athletic performance.[1] Intensive and demanding training sessions over consecutive days and long competitive seasons all challenge athletes' recovery.[1] Fundamental to recovery and performance is sleep, and while 7–9 h of sleep per night is typically recommended for optimal health,[2] it has been speculated that athletes need more than this to optimise performance.[3] However, training and competition schedules, transmeridian travel and pre-competition anxiety may impair sleep[4], and ultimately performance. Optimizing sleep is, therefore, critical for all athletes. This is especially true for professional elite Australian Rules footballers who often report obtaining less than 7 h of sleep per night.[5]

One plausible strategy recommended to improve sleep is dietary intake.[6] A number of studies have investigated the influence of

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sleep restriction, sleep deprivation and shift work on dietary choices, often resulting in more frequent snacking, higher carbohydrate and lower protein intake,[7] however the literature investigating the opposite relationship is limited. Current evidence indicates that nutritional factors including energy intake, macronutrient composition and meal timing, may influence sleep.[7] For example, energy restriction has been found to increase sleep onset latency and reduce slow wave sleep in overweight women.[8] In addition, high carbohydrate (HC) diets have reduced slow wave sleep,[9] while a very low carbohydrate diet was shown to reduce rapid eye movement sleep and increase slow wave sleep.[10] The quality of carbohydrate intake is also important, as poor sleepers were found to consume the highest carbohydrate intake from refined sources including confectionary and sugar sweetened beverages.[11] High protein intakes have been suggested to increase arousal and restlessness,[6,12] while low protein intakes (<16% of total energy) have also been associated with poor sleep quality,[12] possibly suggesting there may be an optimal protein intake for sleep. The type of fat intake has also been shown to influence sleep, with high saturated fat intakes reducing sleep quality,[13] while a diet high in unsaturated fats has been found to improve sleep quality.[14] Although equivocal, the current body of literature suggests potential associations between nutrient intake and sleep, however no studies have been conducted in athletes.

The timing of night-time nutrient intake may also influence sleep. It is a common recommendation to avoid eating a meal within the 2–3 h before bed,[15] despite limited support from scientific evidence. One study has shown a shorter sleep onset when a meal is consumed 4 h prior to bedtime compared to 1 h.[16] This provides some insight on the effects of evening meal consumption on sleep, suggesting that the timing of food before bed could be an important consideration.

While there is evidence that the size, composition, and timing of meals can affect sleep, it remains unclear whether these relationships persist in athletic populations. Therefore, aims of the current study were to investigate the relationship between total daily and evening meal calorie intake, macronutrient composition, evening meal timing and sleep in elite male Australian Football League (AFL) players.

2. Methods

A prospective cohort study design was used to examine dietary intake and sleep/wake behaviour in 36 elite male AFL players (23.5 ± 3.9 y; 86.6 ± 8.1 kg; 189 ± 7 cm) from one professional club. Data collection was conducted for ten days during the preseason period of the 2018 season, which included air travel on the ninth day so that players could participate in a pre-season practice match on the tenth day. The research study was approved by the Deakin University Human Research Ethics Committee (HEAG-H 182_2017) in December 2017.

Nutritional intake was assessed via photo food diaries using the smartphone application, MealLogger Pro (Wellness Foundry, Ashburn, VA), where players took photos and entered details of all food, beverages and supplements ingested. All entries were checked by an accredited sports dietitian and players were followed up daily for any other necessary detail and to account for incorrect reporting. The Australian-specific dietary analysis software program FoodWorks (Version 9, Xyris Software, Canberra, Australia) was used to quantify the dietary data. All data entry into FoodWorks was completed by one trained nutrition researcher to ensure consistency. Dietary variables derived from the data included total energy (kJ, derived from macronutrients, including alcohol), protein (g), fat (g), carbohydrate (g), sugar (g) alcohol (g) and caffeine (mg) content of each meal and daily totals. The same variables were also calculated for "evening" data (defined as intake after 6 pm), comprising of both dinner meal and any further intake before bed. Macronutrient (carbohydrate, sugar, protein, total fat and saturated fat) intakes in g kg^{-1} of body mass were calculated to enable relative comparisons. Timing of the evening meal and evening snack was calculated as the hours between the meal or snack and bedtime, respectively, determined by timestamps on food posts and self-reported bedtime as indicated by sleep diaries.

Players' sleep/wake behaviour was monitored using self-reported sleep diaries administered via Smartabase (Fusion Sport, 2007–2019, Chicago, IL), an online player-management system, in conjunction with wrist activity monitors (Actical Z MiniMitter; Phillips Respironics, Bend, OR). The activity monitors recorded movement in one-minute epochs and data were downloaded using a device-specific interface unit (ActiReader, Phillips Respironics, Bend, OR). Data were then processed using a propriety algorithm set to a medium threshold (<40 counts min^{-1} deemed sleep).[17] This threshold is recommended for team sport athletes,[18] and previously demonstrated a sensitivity (i.e., percentage of sleep epochs correctly detected) of 87.5%, and a specificity (i.e., percentage of wake epochs correctly detected) of 77.1%, when compared with polysomnography.[17] Data recorded by sleep diaries (i.e., bedtime, get-up time) were used to verify, and identify misclassified, sleep/wake states calculated by actigraphy.[19] Players slept at home during data collection; however, 23 players spent the final night of data collection in a single occupancy hotel room after travelling 3 h north (no change in time zone) to participate in a pre-season practice match. Sleep variables extracted from activity monitors and sleep diaries included:

- *Bedtime (hh:mm):* the self-reported clock time at which a participant went to bed to attempt to sleep.
- *Wakeup time (hh:mm):* the self-reported clock time at which a participant woke up and stopped attempting to sleep.
- *Sleep onset latency (SOL; min):* the period of time between bedtime and sleep onset time.
- *Total sleep time (TST; h):* the total amount of sleep obtained during a sleep period.
- *Sleep efficiency (SE; %):* the percentage of time in bed that was spent asleep.
- *Wake after sleep onset (WASO; min):* the total amount of time spent awake during a sleep period.

Player chronotypes (e.g., morning types, evening types or in-between) was considered a possible confounder for the nutrition-sleep relationships and determined via the validated short version Morningness–Eveningness Questionnaire.[20] Internal training load for each football and resistance training session was also considered a potential confounder, quantified using the session RPE method (i.e., duration × RPE; CR-10 scale).[21]

All statistical analyses were conducted using the IBM Statistical Package for Social Sciences Version 24.0 (IBM Corp., Armonk, NY). Results were reported as mean ± standard deviation (SD). Generalized linear mixed models were used to assess the associations between diet [total daily and evening energy (MJ), protein (g and g kg^{-1}), carbohydrate (g and g kg^{-1}), sugar (g and g kg^{-1}) and fat (g and g kg^{-1}) intake and evening meal timing] and sleep [TST, SE, WASO and SOL]. Statistically significant associations ($p < 0.05$) are presented as a beta (β) value, indicating a change in the outcome variable for every 1-unit increase in the predictor variable. Covariates were selected based on their known impacts on sleep. For the different models there were between 170–260 cases included, therefore no more than a total of 15 covariates were included in any model.[22] The covariates included in all analyses were: training load, TST for the previous night (min), total daily alcohol (g) and caffeine (mg) intake, air travel to pre-season practice match (Y/N),

age (years) and Morningness–Eveningness Questionnaire results. Bodyweight was included as a covariate for models using total daily intake. Macronutrients were included as covariates for analyses assessing total daily energy and sleep variables. In models where a macronutrient was used as a predictor, remaining macronutrients were included as covariates.

3. Results

The mean daily intake of energy was 14 ± 4 MJ. The mean daily intake of carbohydrate, protein and fat were 298 ± 117 g (3.4 ± 1.4 g kg^{-1}), 191 ± 66 g (2.2 ± 0.8 g kg^{-1}) and 151 ± 62 g (1.7 ± 0.7 g kg^{-1}), respectively. An overview of average sleep/wake behaviours, evening meal time, training/competition schedule and travel time are shown in Fig. 1. The average bedtime was 22:40 (hh:mm) and wakeup time was 07:50 (hh:mm). The mean TST was 7.9 ± 1.1 h, WASO was 45 ± 20 min, SE was 91 ± 3 % and average SOL was 5 ± 9 min.

The associations between *daily* macronutrient intake and sleep outcomes are shown in Table 1. Every 1-g and 1-g kg^{-1} increase in total daily protein intake was associated with a decrease in SE by 0.01 % (p = 0.007) and 0.7 % (p = 0.006), respectively. Significant associations were also found between total daily energy intake (MJ), total daily protein intake (g and g kg^{-1}), meal timing and WASO (Table 1). Every 1-MJ increase in total daily energy intake was associated with a three minute increase in WASO (p = 0.032). Every 1-g and 1-g kg^{-1} increase in total protein intake was associated with an increase in WASO by 0.04 min (2 s; p = 0.014) and 4 min (p = 0.013), respectively.

The associations between *evening* macronutrient intake and sleep outcomes are shown in Table 2. Every 1-g and 1-g kg^{-1} increase in evening sugar was associated with a decrease in TST by 0.1 min (6 s; p = 0.039) and 5 min (p = 0.027), respectively; an increase in SE by 0.002% (p = 0.015) and 0.2 % (p = 0.021), respectively and a decrease in WASO by 0.012 min (1 s; p = 0.003) and one minute (p = 0.005), respectively. Significant associations were found between evening energy (MJ) and protein (g) and (g kg^{-1}) intakes and SOL (Table 2). Every 1-MJ increase in evening energy intake was associated with a increase in SOL by 5 min (p = 0.011). Every 1-g and 1-g kg^{-1} increase in evening protein intake was associated with a decrease in SOL by 0.03 min (2 s; p = 0.013) and 2 min (p = 0.013), respectively.

Significant associations were found between timing of the main evening meal and TST [β-8.1 (95%CI: 15.8, −0.3), p = 0.042] and WASO [β-2.3 (95%CI: −4.1, −0.4), p = 0.015]. Every additional hour between the main evening meal and bedtime was associated with a decrease in TST by 8 min (p = 0.042) and decrease in WASO by 2 min (p = 0.015). Significant associations were also found between timing of last food or drink (excluding water) [β-6.5 (95%CI: 11.6, −1.3), p = 0.014] and TST. Every additional hour between the last food or drink consumed and bed time was associated with a decrease in TST by 6 min (p = 0.014).

4. Discussion

This is the first study to assess the association between dietary intake, meal timing and sleep in elite athletes. The key findings from this study showed that increasing the time between last food consumption and bedtime, or consuming a higher intake of sugar in the evening were each associated with shorter TST in male AFL players. Further, a higher total daily energy intake and higher evening energy intake was associated with a longer WASO and SOL, respectively. Similarly, a higher total daily protein intake was associated with a longer WASO and reduced SE, while a higher evening protein intake was associated with a shorter SOL.

There was a complex relationship between meal timing and sleep outcomes, as both TST and WASO reduced as the time between final meal consumption and bedtime increased, suggesting sleep was longer, but more disturbed, when meals were consumed close to bedtime. Although it is generally recommended that adults avoid food in the hours before bedtime[15], there is limited evidence to support this in athletic populations. Similar to the current study, one study examining Dutch youth athletes found that a heavy meal within 3 h of bed was associated with increased TST compared with a meal more than 3 h before bed.[23] Given active populations (e.g., athletes) have high metabolic demands,[3] it could be speculated that a longer time between food consumption and sleep may increase feelings of hunger and reduce satiety,[24] which may prompt athletes to wake earlier. In contrast, eating closer to bedtime may induce metabolic disturbance due to digestion, raising core body temperature[25] and possibly leading to more disturbed sleep. Future research should explore relationships between athletes' satiety, appetite-regulating hormones, core body temperature and sleep. However, the magnitude of the association between meal timing and WASO was small (∼2 min decrease for every hour food was consumed away from bedtime), suggesting that the benefit of the increased TST may outweigh the increased WASO. This is an important finding for athletes who often train or compete in the evening, as they may have the confidence to eat later without impacting on sleep and recovery.

A higher sugar intake in the evening was associated with a shorter TST. Although no previous studies have investigated the effect of sugar intake on sleep, one study investigated the effects of high GI on sleep in the general population.[16] Many sugars, in particular refined sugar, are considered to be high GI, meaning that it is digested quickly after consumption, leading to a spike in blood glucose.[16] Unlike the current study, Afaghi et al. showed that consuming a high GI meal either 4 or 1 h before bedtime did not affect TST, however consuming the meal 4 h before bedtime reduced SOL.[16] The average time of evening sugar consumption in the current study was 3 h and 50 min before bedtime, potentially suggesting that the timing of sugar intake may be important. Given the scarce literature, it is difficult to understand the reasoning behind the association between sugar and TST shown in our study, however one possibility is the population group, being elite athletes. Interestingly, one participant in Afaghi's study was an elite athlete who, unlike the other participants, experienced extended SOL following the consumption of a high GI meal prior to bed.[16] Future research should examine whether sugar affects the sleep of athletes and non-athletes differently.

The current findings indicate sleep was slightly more disturbed when daily protein intake increased, shown by a longer WASO and reduced SE. The association between daily protein intake and WASO was small and unlikely to be meaningful. However, AFL players consume high amounts of daily protein to promote recovery and training adaptation (2.4 g kg^{-1} day in the current study), possibly explaining the low SE values (85%) previously reported in AFL players.[5] The impact of protein on sleep may depend on the amino acid profile. Meat-based proteins, which constituted a large proportion of players' protein intake in the current study, may reduce the transport of tryptophan (a precursor of melatonin) into the brain,[26] and reduce sleepiness. However, given this study did not focus on amino acid profiling and melatonin was not measured, the effect of protein type on tryptophan levels is unclear. Further research is required to examine the impact of proteins, specifically the amino acid profile, on athletes' sleep quantity and quality.

Furthermore, an increase in evening protein intake (>6 pm) was associated with a reduced SOL (4-min decrease in SOL for every 1-g kg^{-1} increase). Given the limitations of SOL measurements (i.e., actigraphy tends to underestimate SOL),[18] and that there was no observed relationship between evening protein intake and TST,

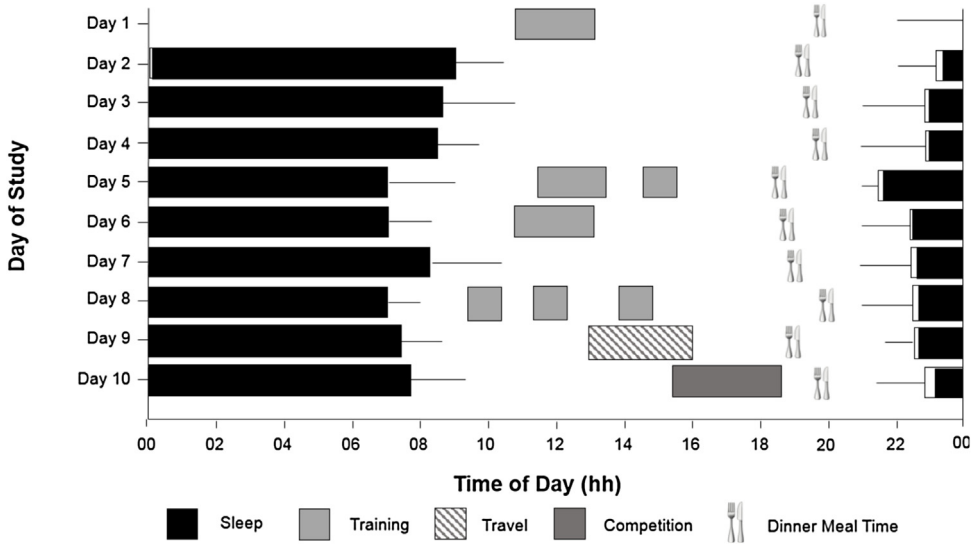

Fig. 1. Average sleep/wake, evening meal timing, training schedule and travel behaviour.

Table 1
Impacts of total daily dietary intake on sleep outcomes[a].

	Total sleep time β (95% CI), p-value	Sleep efficiency β (95% CI), p-value	Wake time β (95% CI), p-value	Sleep onset latency β (95% CI), p-value
Energy[b,f]	−3.5 (−6.6; 13.7), 0.496	−0.3 (−0.7; 0.1), 0.187	2.6 (0.2; 5.0), 0.032*	0.7 (−0.2; 1.6), 0.116
Carbohydrate[c]				
(g)[f]	0.01 (−0.1; 0.1), 0.846	−0.00 (−0.01; 0.00), 0.155	0.01 (−0.01; 0.03), 0.160	−0.01 (−0.01; 0.00), 0.285
(g kg^{-1})	1.5 (−6.0; 9.0), 0.698	−0.2 (−0.5; 0.1), 0.150	1.3 (−0.5; 3.0), 0.156	−0.4 (−1.2; 0.4), 0.383
Sugar (g)[f]	−0.01 (−0.07; 0.05), 0.695	−0.00 (−0.01; 0.00), 0.420	0.02 (−0.01; 0.04), 0.299	0.01 (−0.01; 0.01), 0.365
Sugar (g kg^{-1})	−0.7 (−6.3; 4.9), 0.797	−0.3 (−0.8; 0.3), 0.332	1.7 (−1.1; 4.5), 0.237	0.4 (−0.5; 1.3), 0.361
Protein[d]				
(g)[f]	0.03 (−0.1; 0.2), 0.622	−0.01 (−0.01; −0.00), 0.007*	0.04 (0.01; 0.08), 0.014*	−0.01 (−0.02; 0.02), 0.621
(g kg^{-1})	3.2 (−9.9; 16.4), 0.632	−0.7 (−1.3; −0.2), 0.006*	4.0 (0.8; 7.2), 0.013*	−0.3 (−2.0; 1.4), 0.717
Fat[e]				
(g)[f]	−0.1 (−0.3; 0.1), 0.531	−0.00 (−0.00; 0.00), 0.860	−0.00 (−0.05; 0.04), 0.845	−0.00 (−0.02; 0.02), 0.900
(g kg^{-1})	−5.9 (−25.5; 13.8), 0.558	0.02 (−0.7; 0.7), 0.968	−0.7 (−4.6; 3.2), 0.732	−0.2 (−1.8; 1.5), 0.852
Saturated fat (g)[f]	−0.1 (−0.4; 0.2), 0.682	−0.01 (−0.02; 0.01), 0.491	0.03 (−0.1; 0.1), 0.509	−0.01 (−0.05; 0.04), 0.828
Saturated fat (g kg^{-1})	−5.9 (−35.8; 23.9), 0.696	−0.5 (−1.9; 0.9), 0.472	2.8 (−5.0; 10.7), 0.478	−0.8 (−5.3; 3.7), 0.732

β; Beta score, CI; confidence interval, g; grams, g kg; grams per kilogram of body weight.
* ($p < 0.05$).
[a] Co-variates in all models included age, training load, previous night TST, travel, MEQ, alcohol and caffeine intake; evening meal timing included for evening variables.
[b] Carbohydrate, protein and fat included as co-variates.
[c] Protein and fat included as co-variates.
[d] Carbohydrate and fat included as co-variates.
[e] Carbohydrate and protein included as co-variates.
[f] Bodyweight included as a co-variate.

the protein-SOL relationship observed here should be treated with caution. However, this supports the concept that the amino acid profile of different types of protein may be important for sleep, as dairy-based protein (including whey) has a high Trp/LNAA.[27] This is relevant for the current cohort as whey-based protein in the form of milk and powdered protein was regularly consumed, reported by 56% of players on 57 occasions. Although we did not examine the quantities of meat- vs. dairy-based proteins in the current study, these findings highlight the importance of assessing the amount, type and timing of protein intake on sleep in future research. Future research should explore whether experimentally manipulating night time whey-protein intake significantly and meaningfully improves sleep onset, quality and quantity in athletes with already high protein intakes.

The findings suggest that a higher overall daily and evening energy intakes are associated with disturbed sleep, as shown with longer WASO and SOL, respectively. Studies in the general population have reported an increase in core temperature following the consumption of an energy dense meal in the evening,[28] likely due to the increased metabolic demand for digestion, potentially compromising sleep quality.[25] Thus, in order to meet the high energy requirements of elite athletes whilst not compromising sleep, athletes need to consider the spread of energy intake and food choices so that a higher percentage of energy dense foods is consumed over the day, ideally around training and limit higher intakes in the evening.

The major strengths of this study include the applied, authentic, in-field examination of a well-defined group of elite professional AFL players. However, it should be acknowledged that actigraphy uses movement to infer sleep/wake states in lieu of more accurate methods (i.e., polysomnography) that directly monitor brain activity. Nonetheless, the use of actigraphy to monitor athletes' sleep is valid,[17,18] and necessary in field settings. The smartphone application MealLogger has not been validated against weighed food diaries, however other similar photo based food records have been[29] and MealLogger has been used previously as an effective way to measure food intake in athletic populations.[30] Therefore, the current findings provide useful insights into the interactions

Table 2
Impacts of evening dietary intake on sleep outcomes[a].

	Total sleep time β (95% CI), p-value	Sleep efficiency β (95% CI), p-value	Wake time β (95% CI), p-value	Sleep onset latency β (95% CI), p-value
Energy[b,f]	12.1 (−9.4; 33.6), 0.270	−0.00 (−1.2; 1.2), 0.996	1.8 (−3.6; 7.2), 0.517	5.0 (1.1; 8.8), 0.011*
Carbohydrate[c]				
(g)[f]	−0.04 (−0.2; 0.1), 0.504	−0.00 (−0.01; 0.00), 0.426	0.01 (−0.02; 0.05), 0.360	−0.00 (−0.02; 0.01), 0.847
($g\,kg^{-1}$)	−3.0 (−14.5; 8.4), 0.607	−0.2 (−0.7; 0.3), 0.445	1.3 (−1.5; 4.1), 0.366	−0.02 (−1.4; 1.4), 0.974
Sugar (g)[f]	−0.1 (−0.1; −0.0), 0.039*	0.00 (0.00; 0.00), 0.015*	−0.01 (−0.02; −0.00), 0.003*	−0.00 (−0.01: 0.01), 0.785
Sugar ($g\,kg^{-1}$)	−5.1 (−9.7; −0.6), 0.027*	0.2 (0.0; 0.3), 0.021*	−1.0 (−1.8; −0.3), 0.005*	−0.1 (−0.8; 0.6), 0.806
Protein[d]				
(g)[f]	0.1 (−0.2; 0.3), 0.510	−0.00 (−0.02; 0.01), 0.592	0.04 (−0.03; 0.11), 0.273	−0.03 (−0.05; −0.01), 0.013*
($g\,kg^{-1}$)	6.2 (−15.4; 27.7), 0.575	−0.3 (−1.6; 0.9), 0.632	3.2 (−3.2; 9.6), 0.327	−2.2 (−4.1; −0.4), 0.016*
Fat[e](g)[f]				
(g)[f]	0.01 (−0.4; 0.4), 0.981	−0.01 (−0.02; 0.01), 0.248	0.03 (−0.04; 0.10), 0.351	0.02 (−0.01; 0.05), 0.156
($g\,kg^{-1}$)	1.8 (−34.7; 38.4), 0.921	−0.6 (−1.8; 0.6), 0.324	2.5 (−3.6; 8.7), 0.418	1.8 (−0.8; 4.3), 0.176
Saturated fat (g)[f]	0.2 (−0.3; 0.8), 0.417	−0.00 (−0.03; 0.02), 0.894	0.05 (−0.1; 0.2), 0.563	0.04 (−0.03; 0.12), 0.259
Saturated fat ($g\,kg^{-1}$)	21.7 (−26.9; 70.3), 0.382	−0.1 (−2.3; 2.2), 0.949	3.5 (−10.2; 17.3), 0.613	3.7 (−3.3; 10.6), 0.303

β; Beta score, CI; confidence interval, g; grams, g kg; grams per kilogram of body weight.
* ($p < 0.05$).
[a] Co-variates in all models included age, training load, previous night TST, travel, MEQ, alcohol and caffeine intake; evening meal timing included for evening variables.
[b] Carbohydrate, protein and fat included as co-variates.
[c] Protein and fat included as co-variates.
[d] Carbohydrate and fat included as co-variates.
[e] Carbohydrate and protein included as co-variates.
[f] Bodyweight included as a co-variate.

between diet and sleep outcomes in AFL players. In addition, given the aim of the study was to examine players' night-time sleep, daytime napping was not monitored. Although daytime napping may influence night-time sleep (i.e., by reducing sleep need), it has been shown that team sport athletes rarely nap,[5] and so this limitation is unlikely to affect the findings of the study. Further, the variation in meal duration was not measured, possibly impacting on the recorded hours between eating and bedtime. However, there was slight within-player variation in the time stamp of the evening meal and snack and all players were in similar daily routines, limited the impact of meal duration variations on the study findings. Though the likelihood of inter-club variation (from the same competition) is highly unlikely, it still needs to be acknowledged that recruitment of players all came from the same professional AFL club, limiting the sample size to the number of players in the AFL team. Further, the findings may be specific to male AFL players given the demands and training schedules of the sport and gender differences, potentially limiting the interpretation of findings to other sporting codes, genders and the general adult population.

5. Conclusions

Daily total intake of nutrients did not appear to have a meaningful association with sleep, implying that evening (>6 pm) dietary factors should be the focus when considering the associations with sleep. Specifically, this study found that evening sugar and meal timing had the greatest association with TST, with a higher sugar intake and longer time between eating and bed associated with reduced TST. Evening protein intake reduced SOL, while evening energy intake extended SOL, however given the known limitations with measuring SOL, these findings may need to be treated with caution. Further research will require intervention studies to establish a cause and effect relationship, manipulating specific nutrients and meal timing to assess whether dietary interventions are a viable means of improving sleep outcomes in both elite male and female athletes.

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References

1. Kellmann M. Preventing overtraining in athletes in high-intensity sports and stress/recovery monitoring. *Scand J Med Sci Sports* 2010; 20(2):95–102.
2. Hirshkowitz M, Whiton K, Albert SM et al. National sleep foundation's updated sleep duration recommendations: final report. *Sleep Health* 2015; 1(4):233–243.
3. Bird SP. Sleep, recovery, and athletic performance: a brief review and recommendations. *Strength Cond J* 2013; 35(5):43–47.
4. Roberts SSH, Teo W-P, Warmington SA. Effects of training and competition on the sleep of elite athletes: a systematic review and meta-analysis. *Br J Sports Med* 2019; 53(8):513–522.
5. Lastella M, Roach GD, Sargent C et al. Sleep/wake behaviours of elite athletes from individual and team sports. *Eur J Sport Sci* 2015; 15(2):94–100.
6. Halson SL. Nutrition, sleep and recovery. *Eur J Sport Sci* 2008; 8(2):119–126.
7. Aisbett B, Condo D, Zacharewicz E et al. The impact of shiftwork on skeletal muscle health. *Nutrients* 2017; 9(3):248.
8. Karklin A, Driver HS, Buffenstein R. Restricted energy intake affects nocturnal body temperature and sleep patterns. *Am J Clin Nutr* 1994; 59(2):346–349.
9. Phillips F, Crisp AH, McGuinness B et al. Isocaloric diet changes and electroencephalographic sleep. *Lancet* 1975; 306(7938):723–725.
10. Afaghi A, O'Connor H, Chin Moi C. Acute effects of the very low carbohydrate diet on sleep indices. *Nutr Neurosci* 2008; 11(4):146–154.
11. Katagiri R, Asakura K, Kobayashi S et al. Low intake of vegetables, high intake of confectionary, and unhealthy eating habits are associated with poor sleep quality among middle-aged female Japanese workers. *J Occup Health* 2014; 56(5):359–368.
12. Tanaka E, Yatsuya H, Uemura M et al. Associations of protein, fat, and carbohydrate intakes with insomnia symptoms among middle-aged Japanese workers. *J Epidemiol* 2013; 23(2):132–138.
13. Crispim CA, Zimberg IZ, Gomes Dos Reis B et al. Relationship between food intake and sleep pattern in healthy individuals. *J Clin Sleep Med* 2011; 7(6):659–664.
14. Jaussent I, Dauvilliers Y, Ancelin M-L et al. Insomnia symptoms in older adults: associated factors and gender differences. *Am J Geriatr Psychiatry* 2011; 19(1):88–97.
15. Kamel NS, Gammack JK. Insomnia in the elderly: cause, approach, and treatment. *Am J Med* 2006; 119(6):463–469.
16. Afaghi A, O'Connor H, Chow CM. High-glycemic-index carbohydrate meals shorten sleep onset. *Am J Clin Nutr* 2007; 85(2):426–430.
17. Sargent C, Lastella M, Halson SL et al. The validity of activity monitors for measuring sleep in elite athletes. *J Sci Med Sport* 2016; 19(10):848–853.
18. Fuller KL, Juliff L, Gore CJ et al. Software thresholds alter the bias of actigraphy for monitoring sleep in team-sport athletes. *J Sci Med Sport* 2017; 20(8):756–760.
19. Ancoli-Israel S, Martin JL, Blackwell T et al. The SBSM guide to actigraphy monitoring: clinical and research applications. *Behav Sleep Med* 2015; 13:S4–S38.
20. Adan A, Almirall H. Horne & Östberg morningness-eveningness questionnaire: a reduced scale. *Pers Individ Differ* 1991; 12(3):241–253.
21. Foster JP, Carl H, Kara M et al. Differences in perceptions of training by coaches and athletes. *S Afr J Sports Med* 2001; 8(2):3–7.
22. Field A. *Discovering Statistics Using IBM SPSS Statistics*, 4th ed. London, Sage Publications, 2013.

23. Knufinke M, Nieuwenhuys A, Geurts SAE et al. Self-reported sleep quantity, quality and sleep hygiene in elite athletes. *J Sleep Res* 2018; 27(1):78–85.
24. Karhunen LJ, Juvonen KR, Huotari A et al. Effect of protein, fat, carbohydrate and fibre on gastrointestinal peptide release in humans. *Regul Pept* 2008; 149(1-3):70–78.
25. Campbell SS, Broughton RJ. Rapid decline in body temperature before sleep: fluffing the physiological pillow? *Chronobiol Int* 1994; 11(2):126–131.
26. Markus CR, Jonkman LM, Lammers JHCM et al. Evening intake of alpha-lactalbumin increases plasma tryptophan availability and improves morning alertness and brain measures of attention. *Am J Clin Nutr* 2005; 81(5):1026–1033.
27. Sousa GTD, Lira FS, Rosa JC et al. Dietary whey protein lessens several risk factors for metabolic diseases: a review. *Lipids Health Dis* 2012; 11(1):67–75.
28. Driver HS, Shulman I, Baker FC et al. Energy content of the evening meal alters nocturnal body temperature but not sleep. *Physiol Behav* 1999; 68(1–2):17–23.
29. Olafsdottir AS, Hörnell A, Hedelin M et al. Development and validation of a photographic method to use for dietary assessment in school settings. *PLoS One* 2016; 11(10).
30. Simpson A, Gemming L, Baker D et al. Do image-assisted mobile applications improve dietary habits, knowledge, and behaviours in elite athletes? A pilot study. *Sports* 2017; 5(3):60.

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